

Land and Water Resources Engineering

- ✓ Hydrology
- ✓ Irrigation and Drainage Engineering
- ✓ Soil and Water Conservation Engineering



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PROFESSIONAL REGULATION COMMISSION
PROFESSIONAL REGULATORY BOARD OF AGRICULTURAL AND BIOSYSTEMS ENGINEERING
TABLES OF SPECIFICATION



SUBJECT B. Land and Water Resources Engineering and Allied Subjects (32%)

I. Hydrology The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
	32%	100	Remembering Recall information Recognizing, listing, describing, retrieving, naming, finding	Understanding Explain, Interpret, summarize, examine paraphrase, classify	Applying Using the information in another familiar situation Implementing carrying out, using, executing	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
	2	8						
1. Describe the hydrologic cycle; hydrology in engineering.	0.5	2	2					
2. Describe the principles of hydrometeorology; weather and hydrology; precipitation, evaporation and evapotranspiration; weather instruments.	0.5	2	1	1				
3. Describe the principles of watershed/surface hydrology; runoff, infiltration, seepage and percolation, stream flow, soil erosion and sediment transport.	0.5	2	1	1				
4. Describe the basic theories of aquifer systems and lithography; water quality analysis.	0.5	2	2					

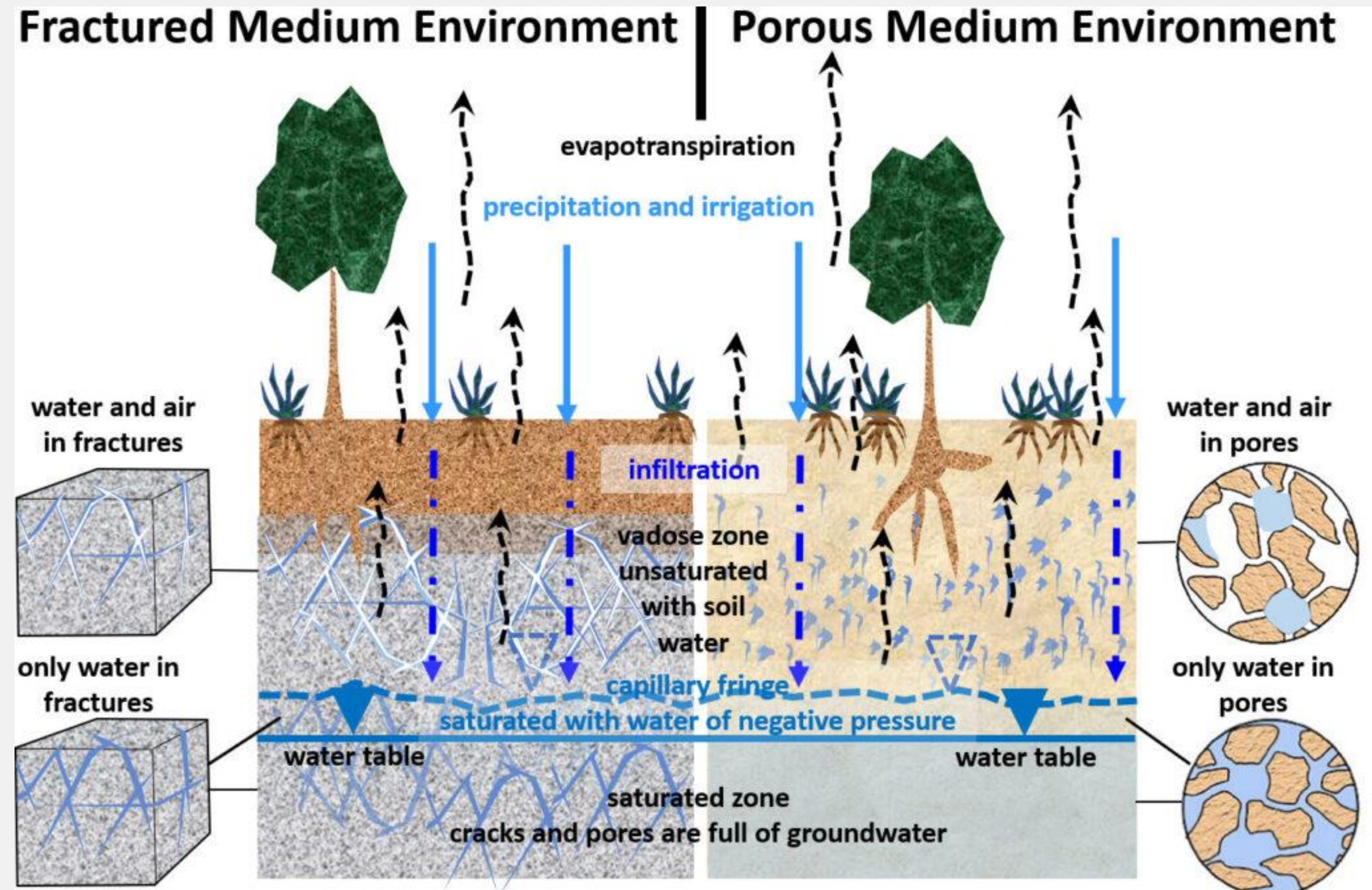
II. Irrigation and Drainage Engineering The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
	32%	100	Remembering Recall information Recognizing, listing, describing, retrieving, naming, finding	Understanding Explain, Interpret, summarize, examine paraphrase, classify	Applying Using the information in another familiar situation Implementing carrying out, using, executing	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
	8	25						
1. Apply theories and practices in leveling, mapping, triangulation, topographic, route, and construction surveys, and global positioning system (GPS).	0.64	2	1		1			
2. Apply the theories and principles of fluid mechanics in the design, construction and management of agricultural and biosystems engineering production systems and facilities.	1.28	4	1	1	2			
3. Apply the basic theories of groundwater hydrology in planning water resource projects.	0.64	2	1		1			
4. Analyze the design, construction, operation and maintenance of hydraulics machinery and facilities (i.e. pumps, and turbines); pump size determination and specifications.	1.28	4				2	2	
5. Plan, design, construct, operate and manage irrigation and drainage systems and facilities; design small canal structures for the conveyance of irrigation or drainage water.	2.88	9		2	1	2	2	2
6. Apply the theories and principles in designing pressurized irrigation systems and facilities.	1.28	4			4			

III. Soil and Water Conservation Engineering The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
	32%	100	Remembering Recall information Recognizing, listing, describing, retrieving, naming, finding	Understanding Explain, Interpret, summarize, examine paraphrase, classify	Applying Using the information in another familiar situation Implementing carrying out, using, executing	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
	6	20						
1. Explain and apply the basic soil-water- plant relations in crop production.	0.6	2		1	1			
2. Apply the principles of hydrometeorology in the planning and design and management of water resources facilities.	0.30	1			1			
3. Apply the principles of watershed hydrology in the planning and design of agricultural and biosystems engineering facilities.	0.30	1			1			
4. Design, construct, manage farm ponds, reservoirs, including earth dams, and spillways.	2.4	8			4	2	1	1
5. Design of soil and water control structures and facilities for stream banks and sloping cultivated agricultural areas.	1.2	4			2		1	1
6. Design of vegetative and lined water conveyance channels.	0.9	3			1		1	1
7. Maintain water conservation works, drainage and erosion control systems.	0.3	1			1			

INFILTRATION

Three Stage Sequence

1. Surface entry
2. Filling up of soil profile storage potential
3. Movement or transmission of water within the soil profile



Factors Affecting Infiltration

1. Fluid
2. Medium
3. Vegetation
4. Cultural practices
5. topography



Measurement of Infiltration



Infiltration Equations

- ✓ **Lewis-Kostiakov**
- ✓ **Horton**
- ✓ **Philip**



Lewis-Kostiakov

$$i = ct^{\alpha}$$

Where I = cumulative infiltration at time t

C, α = constants or parameters



Features of **Lewis-Kostiakov**

L-K equation implies that as $t \rightarrow \infty$, $f = 0$.

L-K will only apply for soil underlain with impervious layers or for small t values in soils with pervious layers.



Horton

$$f = fc + (fo - fc)e^{-kt}$$

f = instantaneous infiltration rate at time *t*

fc = final or constant infiltration rate

fo = initial infiltration rate

$$i = fct + \frac{(fo - fc)}{k} (1 - e^{-kt})$$



Sample Question:

The initial infiltration capacity of a watershed is estimated as 1.5 cm/hr, and the time constant to be 0.35/hr. The equilibrium capacity is estimated as 0.2 cm/hr.

What are the values of f at $t = 10$ min, 30 min, 1 hr, and 6 hr?

What is the total volume of infiltration over the 6 hour time period?

Sample Question:

The initial infiltration capacity of a watershed is estimated as 1.5 cm/hr, and the time constant to be 0.35/hr. The equilibrium capacity is estimated as 0.2 cm/hr.

What are the values of f at $t = 10$ min, 30 min, 1 hr, and 6 hr?

What is the total volume of infiltration over the 6 hour time period?

1.43 cm/hr, 1.29 cm/hr, 1.11 cm/hr, 0.36 cm/hr
4.46 cm

Sample Question:

The following infiltration capacity equation has been derived based on experimental data: $f = 8 + 6e^{(-10.0t)}$ where f is in mm/hr and t is in hr. The cumulative infiltration in mm after 30 minutes is

- a) 4.0 mm
- b) 3.9 mm
- c) 4.6 mm
- d) 8.0 mm

Sample Question:

Use the horton equation to estimate the infiltration rate of a soil after 20 min of infiltration. The initial infiltration rate of soil is 75 mm/h, the steady-state infiltration rate is 20 mm/hr and the constant k is 12.5 per hour.

- a) 20.85
- b) 11.0
- c) 31.2
- d) 23.6

A

Features and use of **Horton Equation**

1. It is **rational** in form and has a better foundation than L-K equation.
2. The equation is very difficult to apply since it contains three parameters that have to be evaluated experimentally.
3. In many instances, Horton equation will not exactly fit infiltration test data.



$$i = St^{1/2} + At$$

S = sorptivity

A = index of gravity



Features and use of **Philip Equation**

1. It has the **strongest theoretical background**.
2. It has been found to be **very accurate** especially for largest values.
3. The parameters **S** and **A** are not empirical factors and can be determined from soil moisture data.



Sample Question 1:

The flow of water from the ground surface into the ground.

- a) Seepage
- b) Infiltration
- c) Penetration
- d) Percolation

B

Sample Question 2:

Run-off is produced when this property of the soil surface is exceeded

- a) Percolation rate
- b) Erodibility
- c) Infiltration rate
- d) Permeability

C

Sample Question 3:

When the rainfall intensity is lower than the infiltration capacity, the actual rate of infiltration is equal to

- a) Infiltration capacity
- b) Saturated hydraulic conductivity
- c) Rainfall intensity
- d) None of the above

Sample Question 4:

It is a device used to measure the rate of water infiltration into soil or other porous media.

- a) Infiltrometer
- b) Evaporimeter
- c) Lysimeter
- d) Hygrometer

Sample Question 5:

According to Robert E. Horton, the equation of infiltration capacity curve, is

- A. $f = f_c (f_o - f_c)e^{kt}$
- B. $f = f_c - (f_o - f_c)e^{-kt}$
- C. $f = f_c + (f_o - f_c)e^{-kt}$
- D. $f = f + (f_o - f_c)e^{kt}$

Sample Question 6:

The infiltration equation based on the exhaustion process is the

- a) Lewis-Kostiakov's
- b) Horton's
- c) Philip's
- d) Darcy's

B

Sample Question 7:

The total observed runoff volume during a storm of 6-hour duration with a uniform intensity of 15 mm/hour is 21.6 MCM. If the area of the basin is 300 km², find the average infiltration rate for the basin.

- a) 1 mm/hour
- b) 3 mm/hour
- c) 8 mm/hour
- d) 10 mm/hour

B

Sample Question:

The rate of vertical water movement through the soil at saturated condition is

- a) infiltration rate
- b) percolation rate
- c) depletion rate
- d) soil permeability

B

Sample Question:

What is the effect of the presence of extensive vegetation in a watershed?

- a) reduce peak runoff rates
- b) increases infiltration capacity
- c) both a and b
- d) none of the above

C

EVAPOTRANSPIRATION

Uses of Evapotranspiration

- ✓ Predicting the irrigation requirements of crops
- ✓ Estimating runoff from river basins
- ✓ Determining the amount of water available from storage reservoir





EVAPOTRANSPIRATION

=

EVAPORATION

+

TRANSPIRATION



Estimating Evapotranspiration

- ✓ Indirect methods
- ✓ Lysimeters
- ✓ Empirical equations



Indirect Method

Pan Evaporation

$$E_{to} = E_{pan} \times k_p$$

k_p = pan coefficient which depends on the type of pan used, pan environment (if the pan is placed in a fallow or cropped area), and the climate (humidity and wind speed)



Lysimeters



POTENTIAL and ACTUAL Evapotranspiration

Potential evapotranspiration is the amount of water that could be evaporated and transpired if there were sufficient water available.

Actual evapotranspiration can be calculated using the equation:

$$E_{tc} = E_{to} \times k_p$$



Empirical Equation (PENMAN METHOD)

Penman Equation

$$ET_o = \frac{\Delta (R_n - G) + \gamma W_f (e_s - e_a)}{\lambda (\Delta + \gamma)}$$

$W_f = 6.43 (1 + 0.53 u_2)$
... for short grass
source Penman (1948)



Sample Question 8:

Is a measuring device which can be used to measure the amount of actual evapotranspiration which is released by plants, usually crops or trees.

- a) Evaporimeter
- b) Lysimeter
- c) Infiltrometer
- d) Barometer

B

Sample Question 9:

What is the Crop water requirement of a plant having a crop coefficient of 0.7 and a Reference Evapotranspiration of 10 mm.

- a) 7 mm
- b) 3.5 mm
- c) 70 mm
- d) 35 mm

Sample Question 10:

A watershed with an area of $2,500 \text{ km}^2$ received 130 cm precipitation. Estimate the annual evapotranspiration if the average rate of flow is $30 \text{ m}^3/\text{s}$.

- a) 37.8 cm
- b) 55.6 cm
- c) 92.2 cm
- d) 101.3 cm

C

Sample Question 11:

A drainage basin is 160 km^2 and receives 112 cm of rainfall per year. The mean annual flow from the basin is $3.1 \text{ m}^3/\text{s}$. What is the average annual evapotranspiration in cm/year ?

- a) 50.9
- b) 54.6
- c) 68.4
- d) 61.1

A

Sample Question 12:

Estimate the evaporation rate from an open surface if the net radiation is 1000 W/m^2 . Given that the temperature of water is 20°C having density of $1,000 \text{ kg/m}^3$, latent heat is $2.45 \times 10^6 \text{ J/kg}$.

- a) 45 mm/day
- b) 35 mm/day
- c) 25 mm/day
- d) 15 mm/day

B

Sample Question 13:

The volume of atmospheric water is $12,900 \text{ km}^3$. The evapotranspiration from land is $72,000 \text{ km}^3 / \text{year}$ and that from ocean is $505,000 \text{ km}^3 / \text{year}$. Estimate the residence time of water molecules in the atmosphere (in days).

- a) 5.5 days
- b) 6.2 days
- c) 8.2 days
- d) 9.6 days

Sample Question 14:

A lake has an area of 10 km^2 . During a specific month the lake evaporation was 90 mm. During the same month the inflow to the lake from a river was on average $1.1 \text{ m}^3/\text{s}$ and the outflow from the lake via another river was on average $1.2 \text{ m}^3/\text{s}$. Also, for the same month a water level increase of 100 mm for the lake was observed. What is the precipitation during the month? Assume: no groundwater flow.

- a) 181.08 mm
- b) 164.08 mm
- c) 215.92 mm
- d) 191.08 mm
- e) None of the above

Sample Question 14:

A lake has an area of 10 km^2 . During a specific month the lake evaporation was 90 mm. During the same month the inflow to the lake from a river was on average $1.1 \text{ m}^3/\text{s}$ and the outflow from the lake via another river was on average $1.2 \text{ m}^3/\text{s}$. Also, for the same month a water level increase of 100 mm for the lake was observed. What is the precipitation during the month? Assume: no groundwater flow.

- a) 181.08 mm
- b) 164.08 mm
- c) 215.92 mm
- d) 191.08 mm
- e) None of the above

$$INFLOW - OUTFLOW = 100 \text{ mm}$$

$$(P + I) - (E + O) = 100 \text{ mm}$$

$$P = 100 + E + O - I$$

C

Sample Question:

Three tanks are installed side by side in a field to measure the evapotranspiration of rice.

Tank A is **bottomless** and is **cropped**.

Tank B is **bottomless** but is **uncropped**.

Tank C has a **bottom** and is **uncropped**.

If the water losses after 2 rainless days are as follows:

Tank A - 14.0 mm

Tank B - 10.0 mm

Tank C - 5.0 mm

The actual crop evapotranspiration is

a) 4.0 mm/day

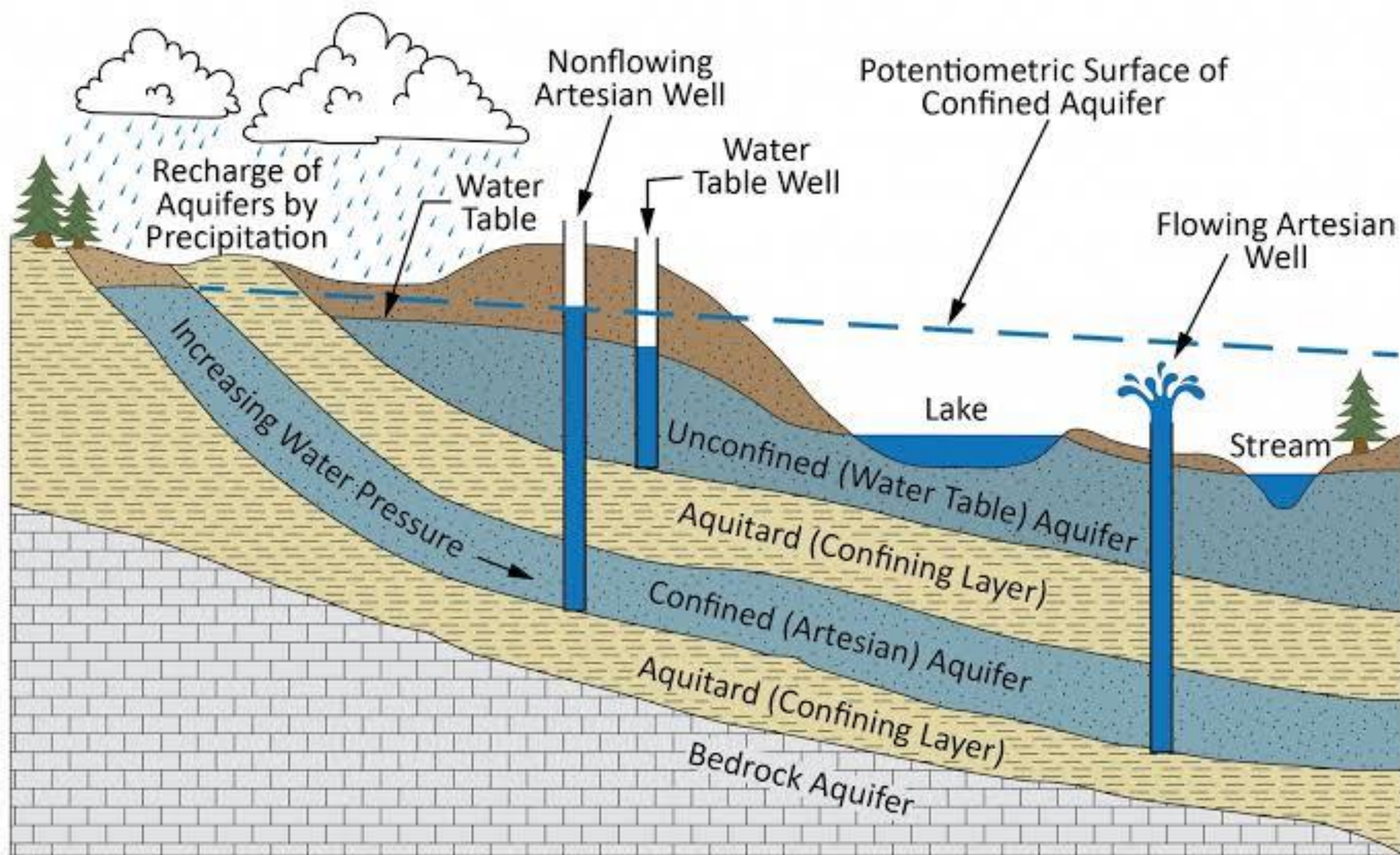
b) 4.5 mm/day

c) 5.0 mm/day

d) 9.0 mm/day

B

GROUNDWATER HYDROLOGY



Definition of Terms

Aquifer – refers to porous geologic formation that is capable of storing and transmitting water at sufficient quantities that permits economic development

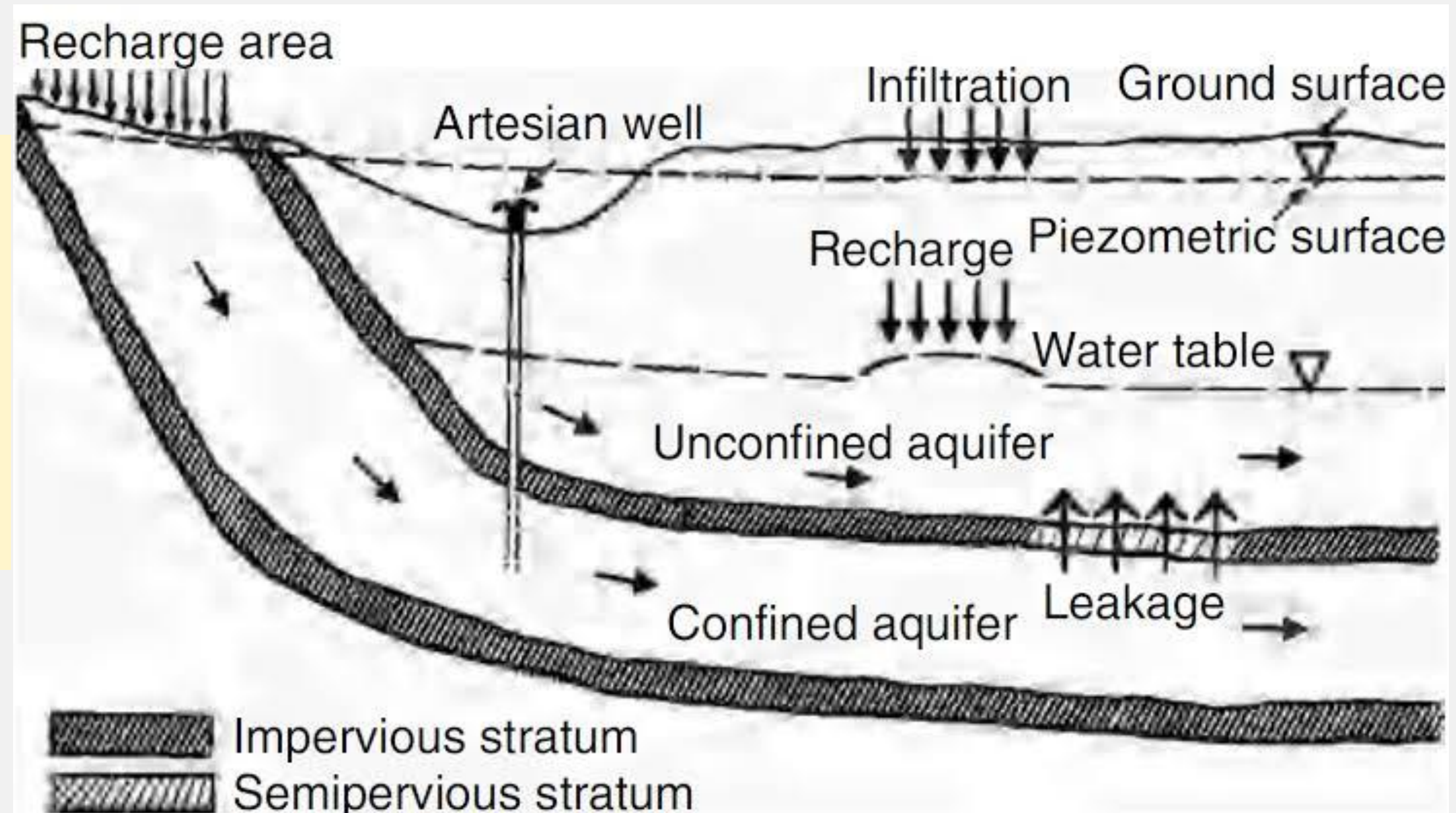
Aquiclude – formation that contains groundwater but not capable of transmitting it at sufficient quantities

Aquifuge – a formation that can neither store nor transmit groundwater

Aquitard – a special formation that can store groundwater but can only transmit it at slow rates

Types of Aquifer

1. Unconfined
2. Confined
3. Semiconfined or leaky



Properties of Aquifer

Hydraulic conductivity – refers to the flow rate per unit area normal to the flow direction resulting from one unit of hydraulic gradient causing the flow

Transmissivity (transmissibility) – an aquifer property which represents its ability to transmit water throughout its entire saturated thickness. It is computed as

$$T = kb$$

where k = average hydraulic conductivity
 b = thickness of the aquifer



Properties of Aquifer

Storage coefficient (storativity) – refers to the volume of water that can be stored in or released from an aquifer per unit decline in head

Specific yield (effective porosity) – refers to the volume of water released from the storage in an unconfined aquifer per unit horizontal area per unit decline in water table.

Specific Retention – refers to the volume of water that remains within the porous portion of the aquifer after releasing the stored volume

Properties of Aquifer

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Transmissivity (transmissibility) – an aquifer property which represents its ability to transmit water throughout its entire saturated thickness. It is computed as

$$T = kb$$

where k = average hydraulic conductivity
 b = thickness of the aquifer

Governing Equations for Groundwater Flow

Darcy's Law

$$Q = kA \frac{dh}{dl}$$

where: Q = groundwater flow rate

k = hydraulic conductivity

A = cross-sectional area normal to the flow direction

dh/dl = hydraulic gradient

Governing Equations for Groundwater Flow

Well Hydraulics

$$Q = -kA \frac{dh}{dr}$$

Where:

dh/dr = hydraulic gradient in radial direction

Governing Equations for Groundwater Flow

Steady Radial Flow in **Unconfined Aquifers**

$$Q = \frac{\pi k (h_2^2 - h_1^2)}{\ln\left(\frac{r_2}{r_1}\right)}$$

Where:

Q = well discharge

k = hydraulic conductivity

h_2 = piezometric head at a radial distance r_2 from the well

h_1 = piezometric head at a radial distance r_1 from the well

r_1 and r_2 are radial distances corresponding to h_1 and h_2

Governing Equations for Groundwater Flow

Steady Radial Flow in Confined Aquifers

$$Q = \frac{2\pi kb(h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)} \text{ or } Q = \frac{2\pi T(h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)}$$

Where:

Q = well discharge

k = hydraulic conductivity

b = average thickness of a confined aquifer

T = transmissivity or transmissibility, $T = kb$

h_2 = piezometric head at a radial distance r_2 from the well

h_1 = piezometric head at a radial distance r_1 from the well

r_1 and r_2 are radial distances corresponding to h_1 and h_2

Sample Question:

A well of radius 30 cm fully penetrating a **water table aquifer**, with hydraulic conductivity of 20 m/day and an initial head above the impervious stratum of 40 m, is pumped till a steady water level on the well is 35 m above the stratum. Assuming that the radius of influence is 600 m and that there are no well losses, estimate the steady state discharge.

Sample Question:

A well of radius 30 cm fully penetrating a **water table aquifer**, with hydraulic conductivity of 20 m/day and an initial head above the impervious stratum of 40 m, is pumped till a steady water level on the well is 35 m above the stratum. Assuming that the radius of influence is 600 m and that there are no well losses, estimate the steady state discharge.

Use the formula for unconfined aquifer:

$$Q = \frac{\pi k (h_2^2 - h_1^2)}{\ln \left(\frac{r_2}{r_1} \right)}$$

Where:

- $k = 20 \text{ m/day}$
- $h_1 = 35 \text{ m}$
- $h_2 = 40 \text{ m}$
- $r_1 = 0.30 \text{ m}$
- $r_2 = 600 \text{ m}$

3,100 m³/day

Sample Question:

A fully penetrating well of diameter 0.3 m abstracts water from a confined aquifer. Compute the discharge when the steady state drawdowns at distances of 10 m and 60 m are respectively 2.4 m and 0.5 m, given the thickness of the aquifer and k as 20 m and 10 m/day respectively.

Use the formula for confined aquifer:

$$Q = \frac{2\pi kb(h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)}$$

Where:

- k = 10 m/day
- b = 20 m
- $h_1 = 0.5$ m
- $h_2 = 2.4$ m
- $r_1 = 10$ m
- $r_2 = 60$ m

1,333 m³/day

Sample Question 15:

It is a geologic formation which transmits water at a rate insufficient to be economically developed for pumping.

- a) Aquifer
- b) Aquiclude
- c) Aquifuge
- d) Aquitard

B

Sample Question 16:

The minimum distance from the center of the lone pumped well wherein no drawdown will be observed is called

- A. Well radius
- B. Radius of interference
- C. Radius of influence
- D. Drawdown curve

C

Sample Question 17:

This results from overlapping radii of influence of neighboring wells

- A. Drawdown
- B. Groundwater decline
- C. Well interference
- D. Drawdown curve

C

Sample Question 18:

Geologic formations arranged in decreasing economic attractiveness of pumping

- a) aquifer, aquifuge, aquitard, aquiclude
- b) aquifer, aquiclude, aquitard, aquifuge
- c) aquifer, aquitard, aquifuge, aquiclude
- d) aquifer, aquitard, aquiclude, aquifuge

D

Sample Question 19:

Volume of water released from storage in an unconfined aquifer per unit horizontal area per unit decline in water table

- a) Specific storage
- b) Specific retention
- c) Specific yield
- d) Transmissivity

C

Sample Question 20:

The field activity performed to determine the hydraulic properties of an aquifer is

- a) Well logging
- b) Pump Testing
- c) Pumping test
- d) Well development

C

Sample Question 21:

The soil property which describes the ability of the soil to transmit water

- a) hydraulic conductivity
- b) percolation rate
- c) matric potential
- d) infiltration rate

A

Sample Question 22:

The other term for gravity well.

- a) confined well
- b) water table well
- c) deep well
- d) well on perched aquifer

B

Sample Question 23:

A fully penetrating well installed in a confined aquifer with a transmissivity of $197 \text{ m}^2/\text{day}$ is pumped at a constant rate. If the drawdowns in two observation wells 8 m and 22 m from the pumped well were observed to be 0.44 m and 1.42 m, respectively, the steady state pump discharge is

- a) 200 gpm
- b) 220 gpm
- c) 240 gpm
- d) 260 gpm
- e) 300 gpm

Sample Question 23:

A fully penetrating well installed in a confined aquifer with a transmissivity of $197 \text{ m}^2/\text{day}$ is pumped at a constant rate. If the drawdowns in two observation wells 8 m and 22 m from the pumped well were observed to be 0.44 m and 1.42 m, respectively, the steady state pump discharge is

Use the formula for confined aquifer:

$$Q = \frac{2\pi T(h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)}$$

- a) 200 gpm
- b) 220 gpm
- c) 240 gpm
- d) 260 gpm
- e) 300 gpm

Where: $T = 197 \text{ m}^2/\text{day}$
 $h_1 = 0.44 \text{ m}$
 $h_2 = 1.42 \text{ m}$
 $r_1 = 8 \text{ m}$
 $r_2 = 22 \text{ m}$

Then convert your answer to gpm.

B

Sample Question 24:

A 5-inch diameter fully penetrating well was installed to extract groundwater from a confined aquifer with a transmissivity of $730 \text{ m}^2/\text{day}$. The radius of influence is 600 ft. If the steady state drawdown in the well is 3m, the well discharge is

- A. 10 Ips
- B. 20 Ips
- C. 30 Ips
- D. 40 Ips

Sample Question 24:

A 5-inch diameter fully penetrating well was installed to extract groundwater from a confined aquifer with a transmissivity of 730 m²/day. The radius of influence is 600 ft. If the steady state drawdown in the well is 3m, the well discharge is

- A. 10 Ips
- B. 20 Ips
- C. 30 Ips
- D. 40 Ips

Use the formula for confined aquifer:

$$Q = \frac{2\pi T(h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)}$$

Where: $T = 730 \text{ m}^2/\text{day}$
 $(h_2 - h_1) = 3 \text{ m}$
 $r_1 = 2.5\text{-inch}$
 $r_2 = 600 \text{ ft}$

Then convert your answer to lps.

B

Sample Question 25:

A 25-ft thick confined aquifer with hydraulic conductivity of 500ft/day was tapped by a 4" shallow tubewell. With a radius of influence of 2000 ft, determine the maximum discharge of the STW in liters per second. Assume an allowable drawdown of 10 ft.

- a) 29.61
- b) 41.15
- c) 27.43
- d) 44.42

Sample Question 25:

A 25-ft thick confined aquifer with hydraulic conductivity of 500ft/day was tapped by a 4" shallow tubewell. With a radius of influence of 2000 ft, determine the maximum discharge of the STW in liters per second. Assume an allowable drawdown of 10 ft.

- a) 29.61
- b) 41.15
- c) 27.43
- d) 44.42

Use the formula for confined aquifer:

$$Q = \frac{2\pi kb(h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)}$$

Where:

- k = 500 ft/day
- b = 25 ft
- $(h_2 - h_1) = 10$ ft
- $r_1 = 2$ -inch
- $r_2 = 2000$ ft

Then convert your answer to lps.

C

Sample Question 26:

A 20-ft thick confined aquifer with hydraulic conductivity of 400 ft/day was tapped by a 4- inch diameter shallow tube well. With a radius of influence of 2,500 ft, determine the maximum discharge of the STW in liters per second. Assume an allowable drawdown of 12 ft?

- a) 22.17
- b) 20.57
- c) 62.71
- d) 25.63

Sample Question 26:

A 20-ft thick confined aquifer with hydraulic conductivity of 400 ft/day was tapped by a 4- inch diameter shallow tube well. With a radius of influence of 2,500 ft, determine the maximum discharge of the STW in liters per second. Assume an allowable drawdown of 12 ft?

- a) 22.17
- b) 20.57
- c) 62.71
- d) 25.63

Use the formula for confined aquifer:

$$Q = \frac{2\pi kb(h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)}$$

Where: $k = 400 \text{ ft/day}$
 $b = 20 \text{ ft}$
 $(h_2 - h_1) = 12 \text{ ft}$
 $r_1 = 2\text{-inch}$
 $r_2 = 2500 \text{ ft}$

Then convert your answer to lps.

B

Sample Question:

It is the volume of water released from storage per unit surface area of aquifer per unit decline in the component of hydraulic head normal to that surface.

- a. Specific Storage
- b. Hydraulic Resistance
- c. Storativity
- d. Leakage

C

***Thank You &
May You All
Win in Life!***

